

Neural Networks project

Optimized Forecasting Models for Gamer Level Identification: A Comparative Analysis of Neural Networks and Machine Learning Algorithms

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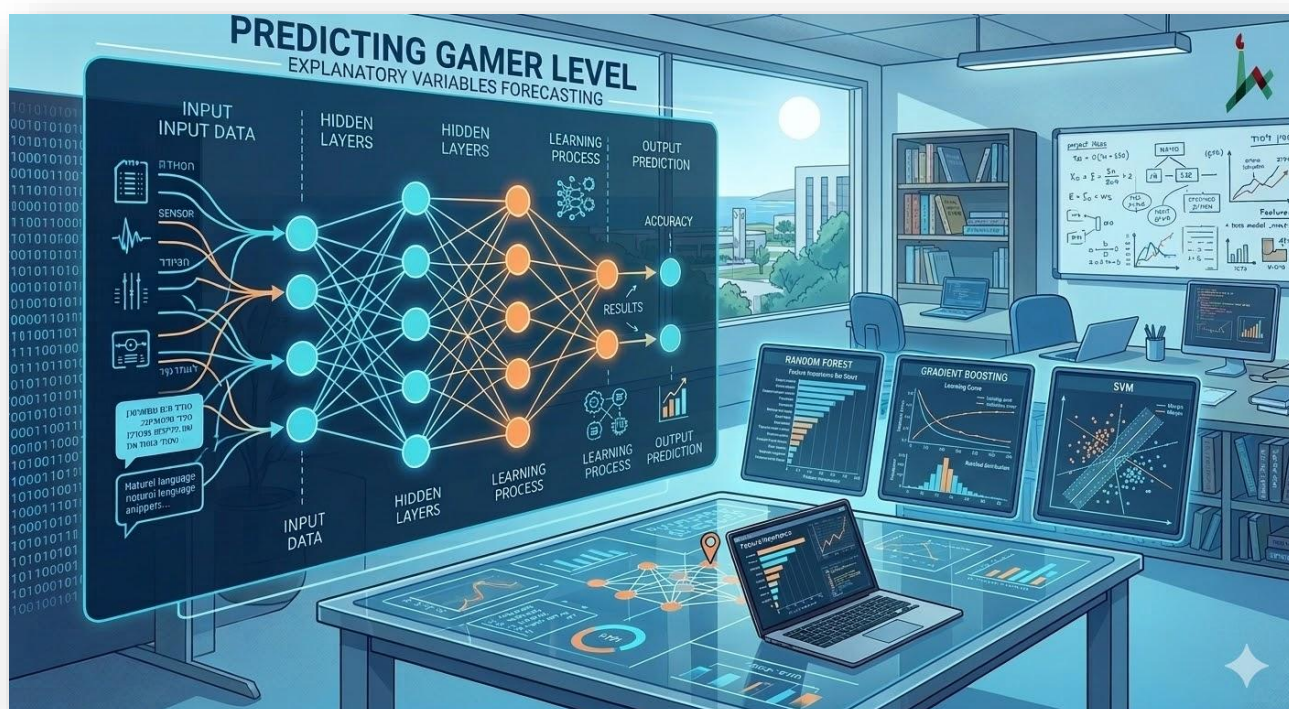


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Introduction

In this project, we apply a range of machine learning algorithms, along with a neural network model, to predict player engagement levels in a gaming environment. The objective is to classify users into three categories—low, medium, and high—based on their behavioral and performance-related features.

The task is formulated as a multi-class classification problem. Various models are implemented and evaluated in order to identify the approach that yields the best predictive performance.

The dataset consists of approximately 40,000 observations and includes both quantitative and qualitative variables describing player characteristics, gameplay behavior, and in-game activity. The full list of variables is presented in the table below.

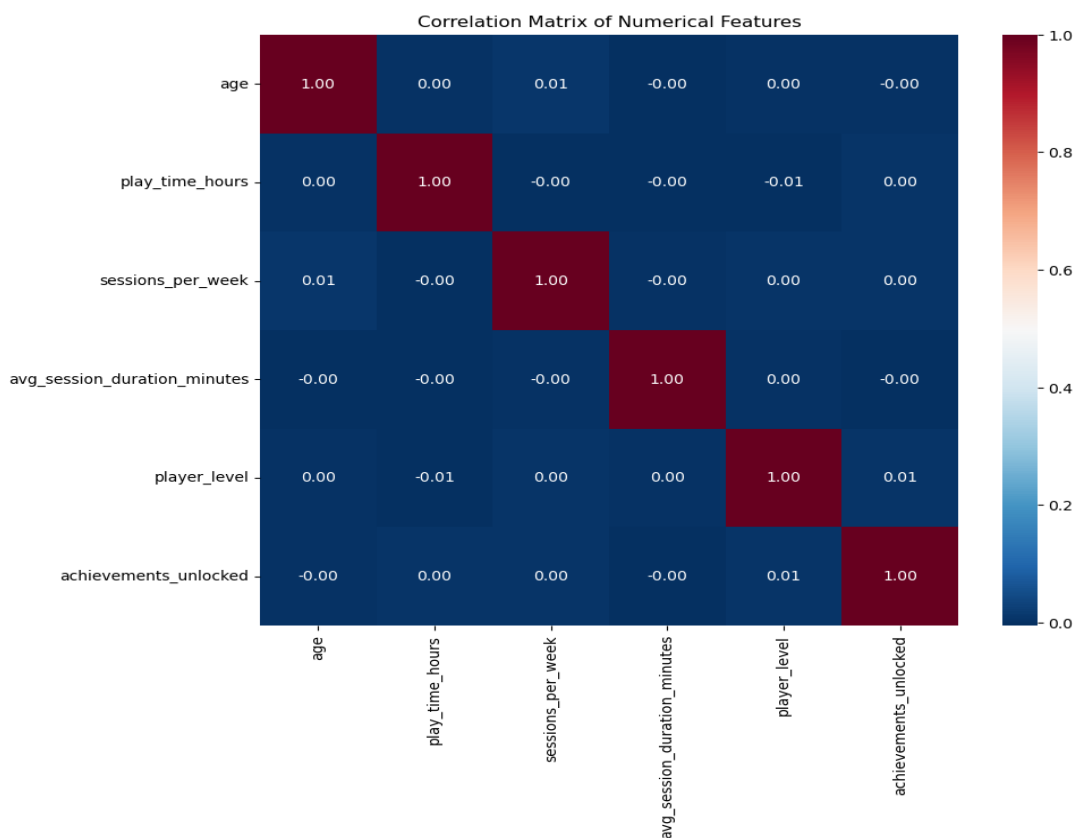
Variable	Quantitative or Qualitative	explanation
Age	Quantitative	Age of the player.
Gender	Qualitative	Gender of the player.
Location	Qualitative	Geographic location of the player.
Game Genre	Qualitative	Genre of the game the player is engaged in.
Play Time Hours	Quantitative	Average hours spent playing per session.
In Game Purchase	Qualitative	Indicates whether the player makes in-game purchase (yes or no)
Game Difficulty	Qualitative	Difficulty level of the game.
Sessions Per Week	Quantitative	Number of gaming sessions per week.
Avg Session Duration Minutes	Quantitative	Average duration of each gaming session in minutes.
Player Level	Quantitative	Current level of the player in the game.
Achievements Unlocked	Quantitative	Number of achievements unlocked by the player.
Engagement Level (target)	Qualitative	Categorized engagement level reflecting player retention ('High', 'Medium', 'Low').

Data Exploration and Quality Check

1. Missing and Duplicate Values

An initial data quality assessment was conducted to ensure the reliability of the dataset. The results indicate that there are no missing values and no duplicate observations, suggesting that the dataset is complete and does not require additional cleaning at this stage.

2. Multicollinearity check

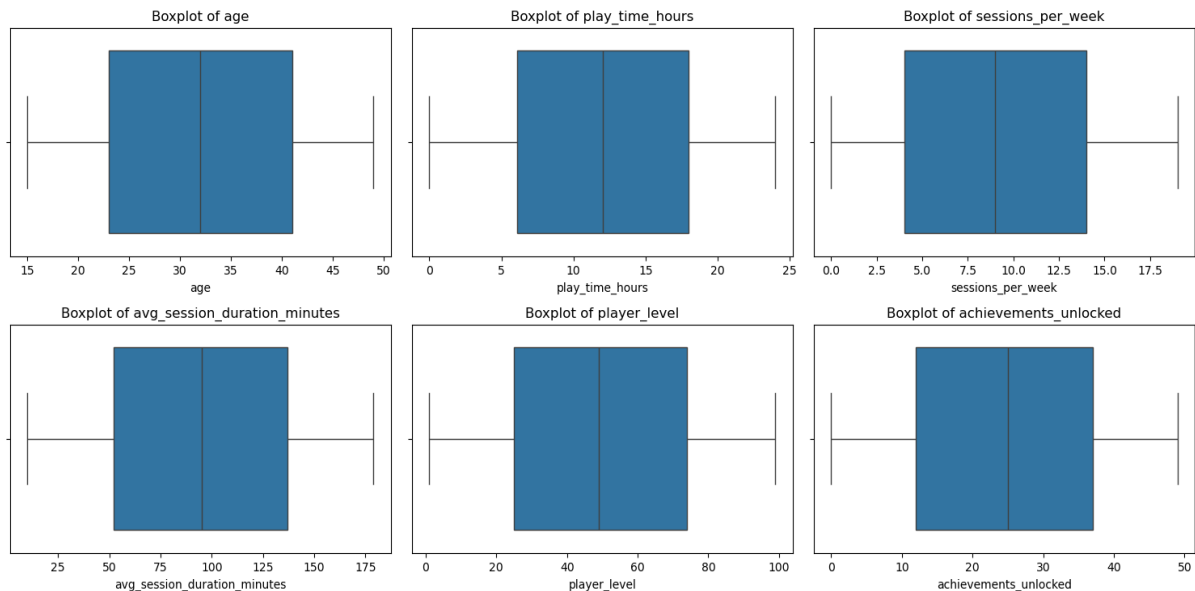


To examine potential multicollinearity among the numerical features, a correlation matrix was computed and visualized using a heatmap. The results show very low correlation between the variables, indicating that multicollinearity is not present in the dataset. This suggests that the features provide distinct information and are suitable for use in the modeling process.

3. Outliers

To identify potential outliers, boxplots were generated for all numerical variables. The visual inspection indicates that the data does not contain significant outliers. Most observations fall within the expected ranges, suggesting a stable distribution across features.

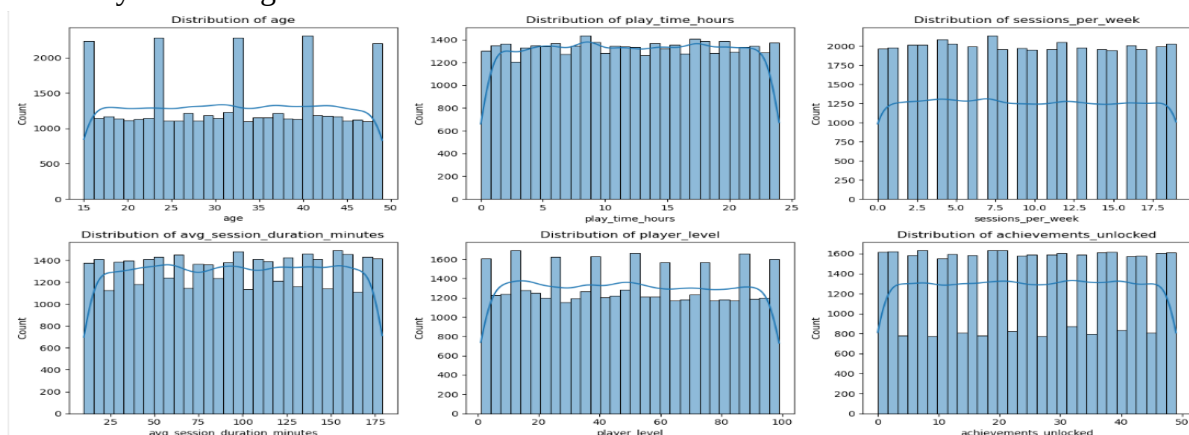
As a result, no outlier treatment was required prior to model implementation.



4. Distribution of the numeric variables:

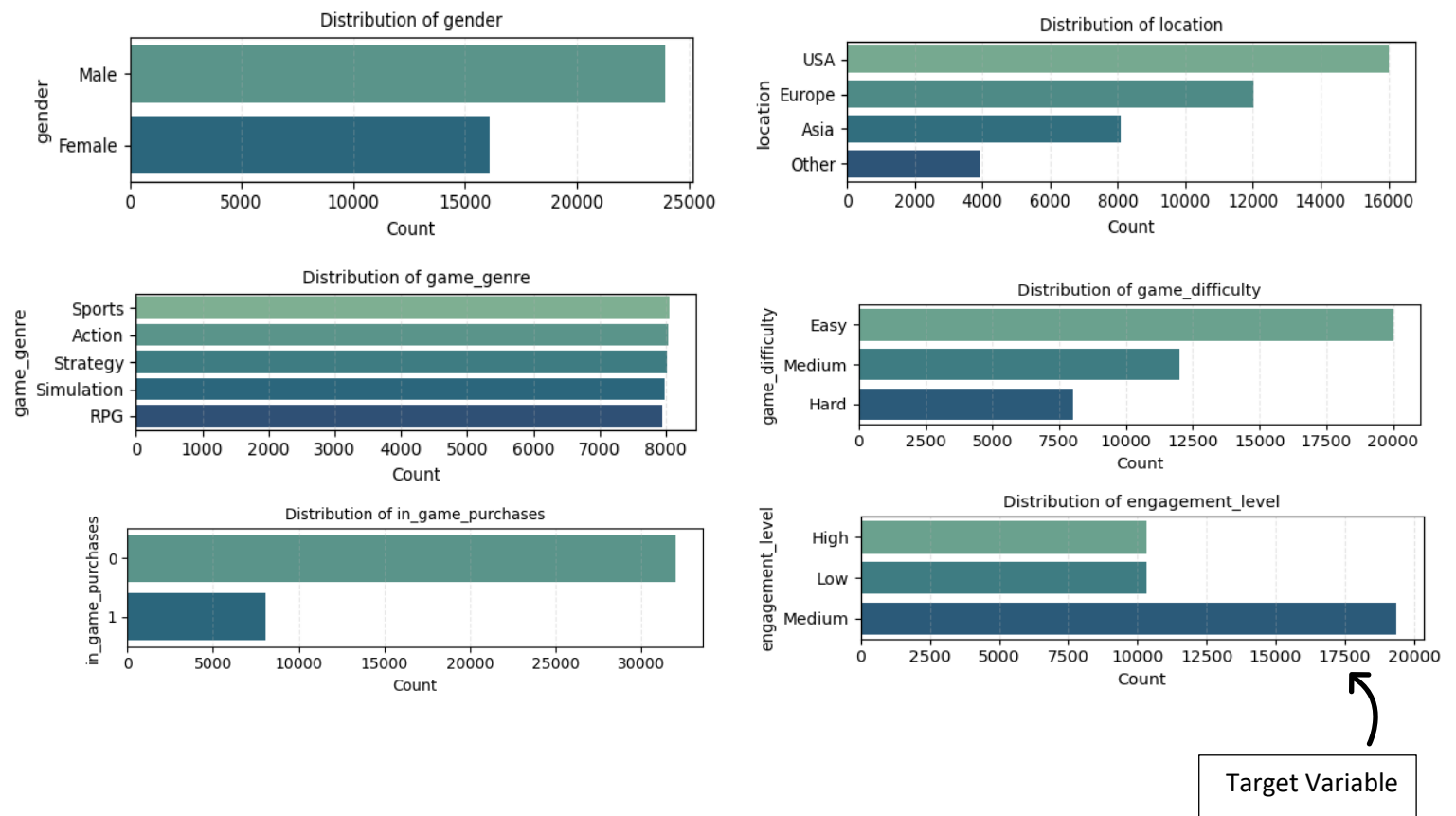
The distribution of the numerical variables was examined using histograms.

The results indicate that most variables follow a relatively uniform distribution rather than a normal distribution. There is no evidence of strong skewness or extreme values, suggesting that the dataset is well-balanced and well-behaved. Consequently, no additional transformations (such as log transformation or scaling for skewness correction) were deemed necessary at this stage.



5. Distribution and Balance of Categorical Variables

The distribution of categorical variables was analyzed to assess class balance across different features. The results indicate varying levels of imbalance. While some variables, such as game genre, appear relatively well-balanced, others—including location, game difficulty, and in-game purchases—exhibit noticeable differences in category frequencies.



Handling of Imbalance in Categorical Variables

No additional preprocessing techniques were applied to address imbalance in the explanatory variables beyond one-hot encoding and feature scaling where required. Although some imbalance is present, it is not severe enough to justify corrective methods such as resampling.

It is important to note that imbalance handling is typically more critical for the target variable rather than for predictor variables. In this case, the class distribution of the target variable was considered acceptable for model training.

Machine Learning Models: Implementation and Comparison

Primary Evaluation Metric: Accuracy

Accuracy was selected as the primary evaluation metric, as the classification task does not involve asymmetric misclassification costs. From an economic perspective, errors across the different engagement levels are assumed to have similar implications, unlike in risk-sensitive applications such as credit scoring. Therefore, accuracy serves as an appropriate and interpretable measure of overall model performance.

Data Preparation and Model Setup

- ✓ Prior to modeling, the dataset underwent preprocessing steps, including one-hot encoding of categorical variables and feature scaling using standardization. The target variable was encoded as a three-class label representing engagement levels (Low, Medium, High).
- ✓ The dataset was split into training and testing sets using an 80/20 ratio, with stratification applied to preserve the class distribution.
- ✓ For modeling purposes, the target variable was encoded numerically, with the levels mapped as follows: Low = 0, Medium = 1, and High = 2.

Multinomial Logistic Regression- Model 1

At this stage, a multinomial Logistic Regression model was implemented to predict the player's engagement level (Low, Medium, High).

Extension from Binary to Multinomial Logistic Regression

Logistic regression is traditionally introduced in its binary form, where the dependent variable consists of two possible outcomes. However, in this project, the target variable includes three categories (Low, Medium, High). Therefore, the model was extended to a multinomial logistic regression framework.

Multinomial logistic regression generalizes the binary case by estimating class probabilities using the SoftMax function. While the core principles remain similar—modeling class probabilities based on explanatory variables—the model enables classification into more than two categories.

This approach is widely used in multi-class classification tasks and is fully supported by standard machine learning libraries such as scikit-learn.

This extension is widely used in multi-class tasks and is fully supported within the scikit-learn

Feature Selection for Logistic Regression

or the multinomial logistic regression model, a backward elimination procedure was applied to retain only statistically significant predictors. Starting with the full set of explanatory variables, features with p-values above 0.08 were iteratively removed, one at a time, until all remaining variables were statistically significant.

This approach improves model interpretability and reduces the inclusion of irrelevant features.

It is important to note that a relatively relaxed significance threshold (0.08) was intentionally chosen. In the context of machine learning, the primary objective is predictive performance rather than strict statistical inference. Therefore, retaining variables with borderline significance may contribute to improved model accuracy, even if they do not meet conventional statistical thresholds such as 0.05.

Most of the selected variables were found to be highly significant ($p < 0.001$), indicating a strong relationship with the target variable. The variable representing in-game purchases showed weaker statistical significance but was retained due to its potential contribution to predictive performance.

The statistical significance of the selected features is summarized in the table below.

Variables	P-value
In Game Purchase	($p \approx 0.485 / 0.06$)
Sessions Per Week	$p < 0.001$
Avg Session Duration Minutes	$p < 0.001$
Player Level	$p < 0.001$
Achievements Unlocked	$p < 0.001$

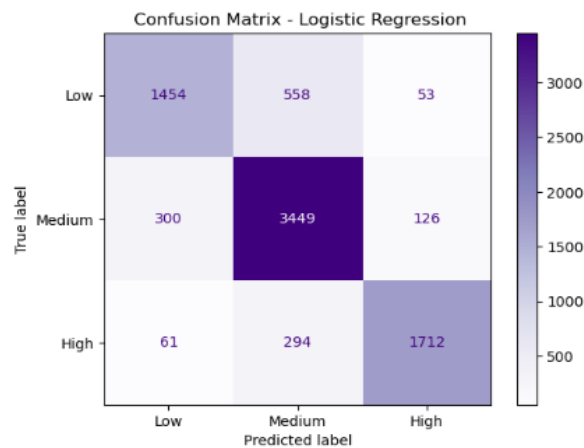
- * The p-values reported for the in-game purchase variable reflect its significance across different outcome categories in the multinomial model. Specifically, the variable was not statistically significant for one category ($p \approx 0.485$) and only marginally significant for another ($p \approx 0.06$), as significance in multinomial logistic regression is evaluated relative to a reference category and may vary across classes.

Results - Logistic Regression

The performance of the Multinomial Logistic Regression model was evaluated using accuracy, confusion matrix, and class-level metrics.

	Precision	Recall	F1-score
0 (low)	0.80	0.70	0.75
1 (medium)	0.80	0.89	0.84
2 (high)	0.91	0.83	0.86
accuracy			0.826

The multinomial logistic regression model achieved an accuracy of 0.826, indicating a strong overall predictive performance.



The confusion matrix provides a detailed view of the model's classification performance across the three engagement levels. The model demonstrates high accuracy in correctly identifying the "Medium" and "High" levels, with a large number of observations correctly classified along the diagonal.

However, some misclassification is observed, particularly between adjacent levels. For example, a notable number of "Low" level observations were misclassified as "Medium," and similarly, some confusion exists between "Medium" and "High" levels.

This pattern suggests that the model captures the overall structure of the data well, but has difficulty distinguishing between neighboring levels, which is expected given the ordinal nature of the target variable.

Logistic Regression - Feature Selection Based on Logistic Regression -results

The results

	Precision	Recall	F1-score
0 (low)	0.80	0.70	0.75
1 (medium)	0.80	0.89	0.84
2 (high)	0.91	0.83	0.86
accuracy			0.825

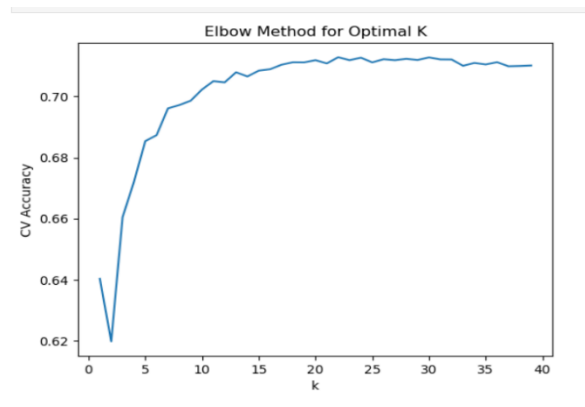
Overfitting check:

- **Using all features**, the model achieved a training accuracy of 0.823 and a test accuracy of 0.826, resulting in a negligible accuracy gap, indicating good generalization with no substantial overfitting.
- **After feature filtering**, the model achieved a training accuracy of 0.824 and a slightly lower test accuracy of 0.826, with similarly stable performance.

This suggests that feature filtering had no meaningful impact on the logistic regression model, which remained stable and well-generalized in both cases.

K-Nearest Neighbors (KNN)- Model 2

The KNN model was implemented with feature scaling and tuned using cross-validation.



An elbow-style analysis guided the selection of the search range, followed by Grid Search ($k = 1-40$), resulting in an optimal value of $k = 32$.

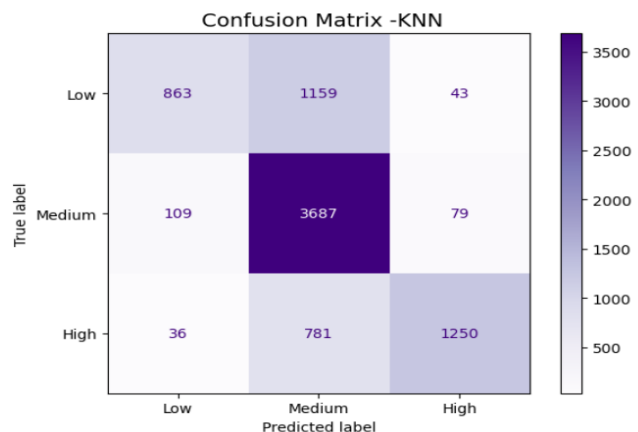
Results- KNN

The performance of the KNN model was evaluated using accuracy, confusion matrix, and class-level metrics.

	Precision	Recall	F1-score
0 (low)	0.86	0.42	0.56
1 (medium)	0.66	0.95	0.78
2 (high)	0.91	0.6	0.73
accuracy			0.72

The K-Nearest Neighbors (KNN) model achieved a test accuracy of **0.724**, which is lower than the performance obtained by the multinomial logistic regression model.

A closer examination of the confusion matrix reveals that the model performs well in identifying the "Medium" level, achieving a high recall of 0.95. This indicates that most observations belonging to the "Medium" class were correctly classified.



However, the model struggles more with the "Low" and "High" levels. In particular, the recall for the "Low" level is relatively low (0.42), suggesting that many "Low" observations were misclassified, primarily as "Medium." Similarly, some confusion is observed between the "High" and "Medium" levels.

Overall, the results suggest that the KNN model tends to favor the central class ("Medium"), which may reflect the structure of the data or overlapping feature distributions between adjacent levels.

While the model captures general patterns, it is less effective at distinguishing between boundary classes compared to the multinomial logistic regression model.

KNN- Feature Selection Based on Logistic Regression

Following the multinomial logistic regression analysis, non-significant variables were identified and removed. The KNN model was then re-trained using the reduced feature set in order to evaluate the impact of feature selection on model performance.

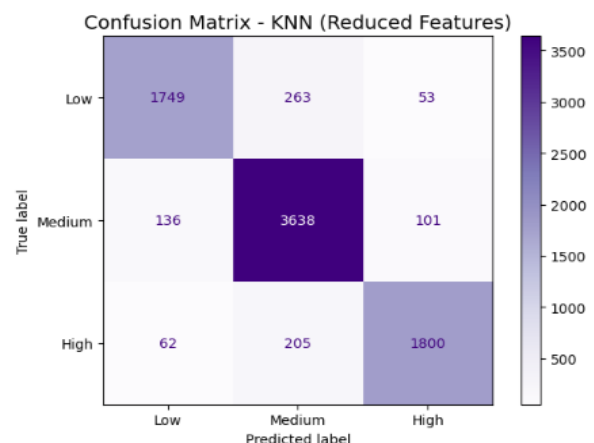
Results – KNN (After Feature Selection)

	Precision	Recall	F1-score
0 (low)	0.90	0.85	0.87
1 (medium)	0.89	0.94	0.91
2 (high)	0.92	0.87	0.90
accuracy			0.897

updated model achieved a significantly improved accuracy of **0.897**, compared to 0.724 obtained using the full set of variables.

This improvement suggests that the removal of irrelevant or noisy features enhanced the model's ability to distinguish between classes. By eliminating less informative variables, the KNN algorithm was better able to identify meaningful patterns in the data.

Notably, the confusion between classes was reduced, indicating improved classification performance across all engagement levels.



Overfitting check:

To assess potential overfitting, the model's performance was also evaluated on the training set.

- **For the model using all features**, the training accuracy (0.746) was slightly higher than the test accuracy (0.724), resulting in a small accuracy gap of 0.021, indicating no substantial overfitting.
- **After removing statistically non-significant variables**, both training and test accuracies improved (0.906 and 0.898, respectively), while the accuracy gap decreased to 0.009.

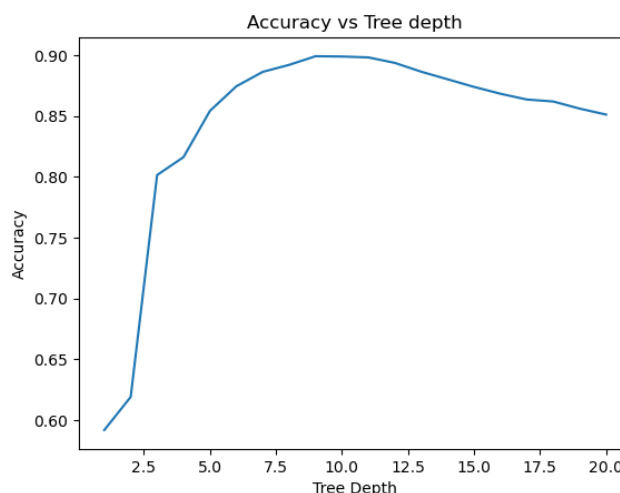
This suggests that feature filtering not only improved predictive performance but also enhanced the model's generalization ability.

Decision Tree- Model 3

The Decision Tree model was implemented and tuned using cross-validation to identify optimal hyperparameters.

The Decision Tree model was first evaluated using a preliminary analysis of model performance across different tree depths. Based on this analysis, an optimal depth of 9 was identified, balancing model complexity and predictive accuracy.

A Grid Search was then conducted to further optimize parameters, including maximum depth and node constraints.

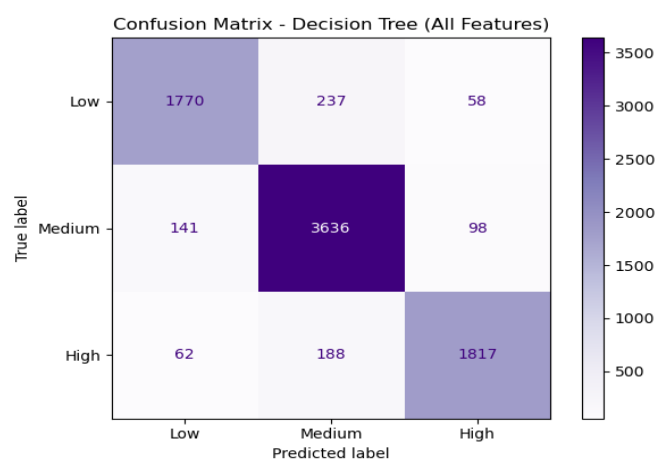


Results

	Precision	Recall	F1-score
0 (low)	0.90	0.86	0.88
1 (medium)	0.90	0.94	0.92
2 (high)	0.92	0.88	0.90
accuracy			0.902

The final model achieved a test accuracy of **0.902**, outperforming both the multinomial logistic regression and KNN models.

The classification report indicates balanced performance across all levels, with precision, recall, and F1-scores close to 0.90. The confusion matrix confirms that the model effectively distinguishes between both central and boundary classes.



The confusion matrix and classification report show balanced performance across all three levels, with high precision, recall, and F1-scores (all close to 0.90). This suggests that the model is able to effectively distinguish between the different engagement levels, including both central and boundary.

Decision Tree - Feature Selection Based on Logistic Regression -results

After removing statistically non-significant variables, the Decision Tree model was re-trained, leading to a slight improvement in test accuracy from 0.902 **to 0.904**.

The results

	Precision	Recall	F1-score
0 (low)	0.90	0.85	0.88
1 (medium)	0.90	0.94	0.92
2 (high)	0.92	0.88	0.90
accuracy			0.904

Overfitting check:

- **Using all features**, the model achieved a training accuracy of 0.920 and a test accuracy of 0.902, resulting in a small accuracy gap of 0.018. This indicates good generalization with no substantial overfitting.
- **After feature filtering**, the model achieved a training accuracy of 0.920 and a slightly improved test accuracy of 0.904, with a similar accuracy gap.

This suggests that feature filtering led to a minor improvement in predictive performance while maintaining stable generalization.

Random Forest- Model 4

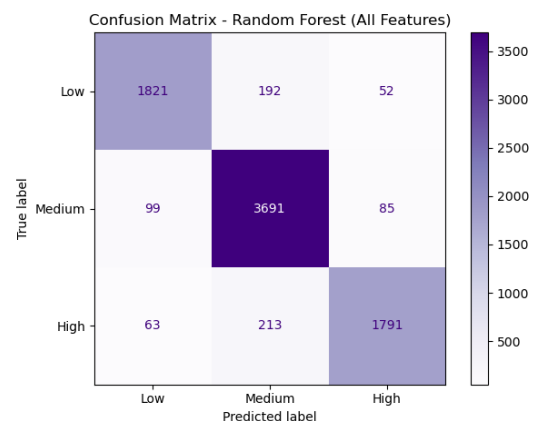
The Random Forest model was implemented and tuned using Grid Search with cross-validation. The hyperparameters included the number of trees, maximum depth, and minimum samples per leaf.

Results

	Precision	Recall	F1-score
0 (low)	0.92	0.88	0.90
1 (medium)	0.90	0.95	0.93
2 (high)	0.93	0.87	0.90
accuracy			0.912

The final model achieved a test accuracy of 0.912, representing the highest performance among the models evaluated so far.

The classification report and confusion matrix indicate strong and balanced performance across all levels, with precision, recall, and F1-scores all close to or above 0.90. This suggests that the model effectively distinguishes between the different engagement levels.



However, an evaluation of training performance reveals a training accuracy of 1.00, resulting in an accuracy gap of approximately 0.088 compared to the test set. This gap suggests potential overfitting, indicating that the model may be capturing noise in the training data in addition to underlying patterns.

Despite this, the model maintains strong generalization performance on the test set, making it one of the best-performing models in this analysis.

Random Forest - Feature Selection Based on Logistic Regression

After removing statistically non-significant variables, the Random Forest model was re-trained using the reduced feature set.

	Precision	Recall	F1-score
0 (low)	0.92	0.90	0.91
1 (medium)	0.92	0.95	0.94
2 (high)	0.93	0.89	0.91
accuracy			0.921

Overfitting check:

- **Using all features**, the model achieved a perfect training accuracy of 1.000 and a test accuracy of 0.912. This large gap suggests potential overfitting, meaning the model may have learned patterns specific to the training data that do not generalize well to unseen data.
- **After feature filtering**, the model achieved a training accuracy of 0.939 and a test accuracy of 0.922, resulting in a much smaller accuracy gap of 0.017.

This indicates improved generalization, as the model maintains high predictive performance while reducing the risk of overfitting.

Gradient Boosting- Model 5

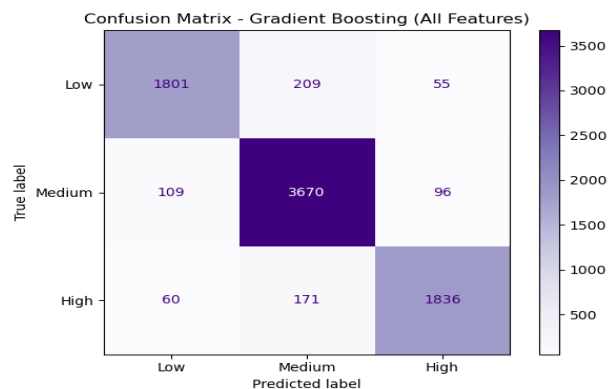
The Gradient Boosting model was implemented and tuned using Grid Search with cross-validation. The hyperparameters included the number of estimators, learning rate, and tree depth.

Results

	Precision	Recall	F1-score
0 (low)	0.91	0.87	0.89
1 (medium)	0.91	0.95	0.93
2 (high)	0.92	0.89	0.91
accuracy			0.913

The final model achieved a test accuracy of 0.913, indicating strong predictive performance, comparable to the Random Forest model.

The classification report and confusion matrix show balanced performance across all levels, with precision, recall, and F1-scores all close to 0.90. This suggests that the model effectively distinguishes between the different engagement levels.



Gradient Boosting - Feature Selection Based on Logistic Regression -results

	Precision	Recall	F1-score
0 (low)	0.91	0.88	0.90
1 (medium)	0.91	0.95	0.93
2 (high)	0.93	0.89	0.91
accuracy			0.915

Overfitting check:

- **Using all features**, the model achieved a training accuracy of 0.920 and a test accuracy of 0.913, resulting in a small accuracy gap of approximately 0.007. This indicates good generalization with no substantial overfitting.
- **After feature filtering**, the model achieved a training accuracy of 0.920 and a slightly improved test accuracy of 0.915, with a similarly small accuracy gap.

This suggests that feature filtering led to a modest improvement in predictive performance while maintaining stable generalization

AdaBoost- Model 6

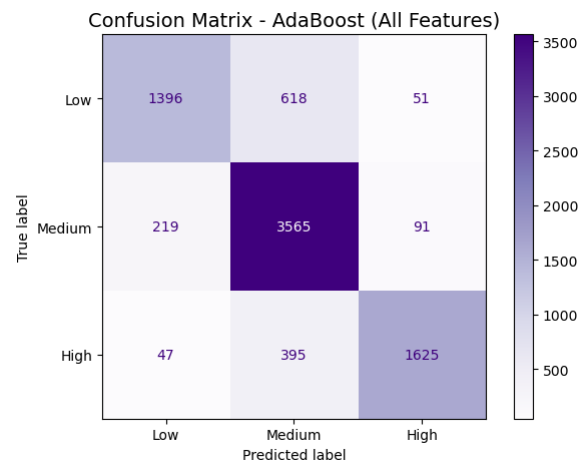
The AdaBoost model was implemented and tuned using Grid Search with cross-validation. The hyperparameters included the number of estimators and the learning rate.

Results

	Precision	Recall	F1-score
0 (low)	0.84	0.68	0.75
1 (medium)	0.78	0.92	0.84
2 (high)	0.92	0.79	0.85
accuracy			0.823

The final model achieved a test accuracy of 0.823, indicating moderate predictive performance compared to the other models.

The classification report and confusion matrix show that the model performs well in identifying the "Medium" level, with a relatively high recall (0.92). However, performance is weaker for the "Low" and "High" levels, with some misclassification toward the central class.



AdaBoost - Feature Selection Based on Logistic Regression -results

	Precision	Recall	F1-score
0 (low)	0.84	0.67	0.75
1 (medium)	0.78	0.92	0.84
2 (high)	0.92	0.79	0.85
accuracy			0.822

Overfitting check:

- **Using all features**, the model achieved a training accuracy of 0.821 and a test accuracy of 0.823, resulting in a negligible accuracy gap, indicating strong generalization with no overfitting.
- **After feature filtering**, the model achieved similar performance, with a training accuracy of 0.821 and a test accuracy of 0.822.

This suggests that feature filtering had no significant impact on model performance, and the model remains stable and well-generalized in both cases.

XGBoost - Model 7

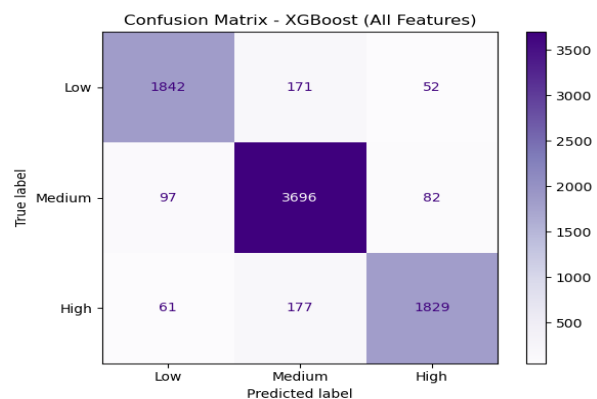
The XGBoost model was implemented and tuned using Grid Search with cross-validation. The hyperparameters included the number of estimators, learning rate, tree depth, and sampling parameters.

Results

	Precision	Recall	F1-score
0 (low)	0.92	0.89	0.91
1 (medium)	0.91	0.95	0.93
2 (high)	0.93	0.88	0.91
accuracy			0.92

The final model achieved a test accuracy of 0.9197, indicating strong predictive performance, comparable to the Random Forest and Gradient Boosting models.

The classification report and confusion matrix show balanced performance across all levels, with precision, recall, and F1-scores all around 0.90. This suggests that the model effectively distinguishes between the different engagement levels.



XGBoost - Feature Selection Based on Logistic Regression -results

	Precision	Recall	F1-score
0 (low)	0.92	0.89	0.91
1 (medium)	0.92	0.95	0.93
2 (high)	0.93	0.89	0.91
accuracy			0.921

Overfitting check:

- **Using all features**, the model achieved a training accuracy of 0.936 and a test accuracy of 0.920, resulting in a small accuracy gap of approximately 0.016. This indicates good generalization with no substantial overfitting.
- **After feature filtering**, the model achieved a training accuracy of 0.933 and a slightly improved test accuracy of 0.921, with a similarly small accuracy gap.

This suggests that feature filtering led to a minor improvement in predictive performance while maintaining stable generalization.

SVM - Model 8

The Support Vector Machine (SVM) model was implemented and tuned using Grid Search with cross-validation. Different kernel functions were evaluated, including linear, polynomial, and radial basis function (RBF), along with hyperparameters such as the regularization parameter (C), degree (for polynomial), and gamma.

Results

linear kernel

	Precision	Recall	F1-score
0 (low)	0.79	0.71	0.75
1 (medium)	0.81	0.87	0.84
2 (high)	0.89	0.84	0.86
accuracy			0.824

polynomial kernel

	Precision	Recall	F1-score
0 (low)	0.90	0.79	0.84
1 (medium)	0.85	0.93	0.89
2 (high)	0.89	0.85	0.87
accuracy			0.872

RBF kernel

	Precision	Recall	F1-score
0 (low)	0.89	0.84	0.86
1 (medium)	0.88	0.94	0.91
2 (high)	0.92	0.86	0.89
accuracy			0.892

The results indicate that model performance varies significantly across different kernel choices.

The linear kernel achieved a test accuracy of 0.824, indicating moderate performance, with almost identical training and test accuracy, suggesting excellent generalization but limited flexibility in capturing complex patterns.

The polynomial kernel improved performance, achieving a test accuracy of 0.872, reflecting its ability to model non-linear relationships.

The RBF kernel achieved the best performance among the SVM variants, with a test accuracy of 0.892, demonstrating strong predictive capability and a good balance between model flexibility and generalization.

Impact of Feature Filtering on SVM Performance

To further evaluate the impact of feature filtering, the SVM model was re-trained using the reduced feature set derived from the logistic regression analysis.

The results show consistent improvement across all kernel types. The linear kernel remained relatively stable, while the polynomial kernel showed moderate improvement. The most notable enhancement was observed for the RBF kernel, where test accuracy increased from 0.892 to 0.922.

RBF kernel (After removing non-significant variables)

	Precision	Recall	F1-score
0 (low)	0.93	0.89	0.91
1 (medium)	0.92	0.96	0.94
2 (high)	0.93	0.89	0.91
accuracy			0.922

Overfitting check:

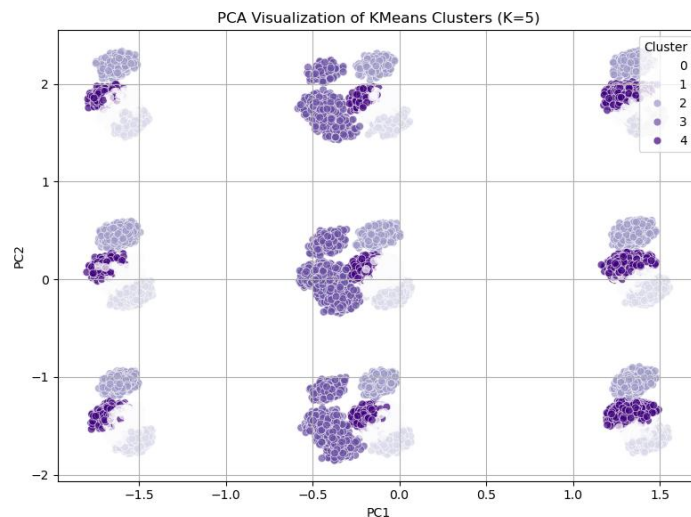
The accuracy gap between training and test performance remained very small across all kernels, indicating strong generalization and no substantial overfitting.

These findings highlight the effectiveness of feature selection in improving SVM performance, particularly for non-linear kernels that are more sensitive to feature space quality.

K-Means Clustering - Model 9

The K-Means clustering algorithm was applied as an unsupervised learning approach to explore the underlying structure of the data.

To determine the optimal number of clusters (k), the silhouette score was evaluated across different values. The highest score was obtained at $k = 5$, indicating that this number of clusters provides the best separation in the feature space.



A PCA-based visualization was used to project the clusters into two dimensions, allowing for visual inspection of cluster separation. While some clustering structure is observed, there is noticeable overlap between clusters, suggesting that the separation between groups is not very strong.

Cluster vs True Target:

True Engagement Level	0	1	2
Cluster			
0	2945	5428	2937
1	1825	3526	1875
2	1881	3485	1799
3	1036	1889	1010
4	2637	5046	2715

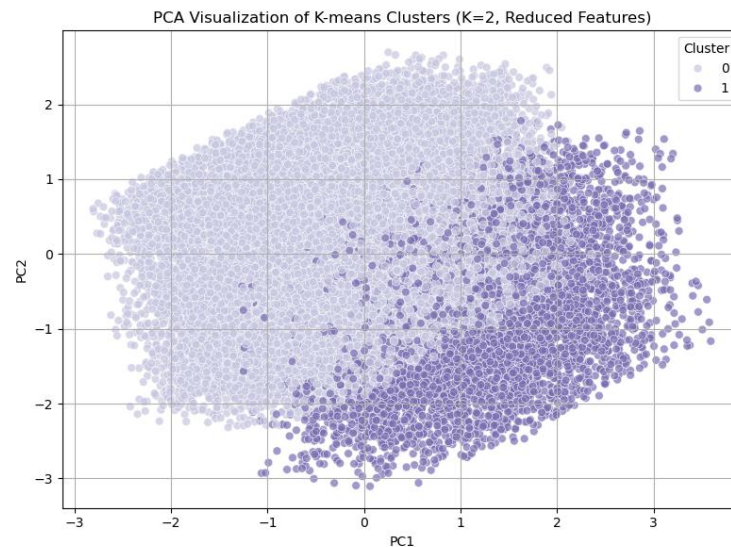
To further evaluate clustering quality, the relationship between cluster assignments and the true engagement levels was examined. The results show that clusters do not perfectly align with the target classes, with each cluster containing a mixture of different engagement levels.

Overall, the K-Means model captures some structure in the data but is less effective at clearly separating the predefined engagement levels compared to supervised learning models.

K-Means After Feature Filtering- Results

To further evaluate the impact of feature filtering, the K-Means model was re-applied using the reduced feature set derived from the logistic regression analysis.

The results show a substantial improvement in clustering quality, with the silhouette score increasing from 0.125 to 0.279. In addition, the optimal number of clusters decreased from $k = 5$ to $k = 2$, indicating a clearer and more distinct underlying structure in the data.



The PCA visualization supports this finding, showing improved separation between clusters compared to the model using all features.

Cluster vs True Target:

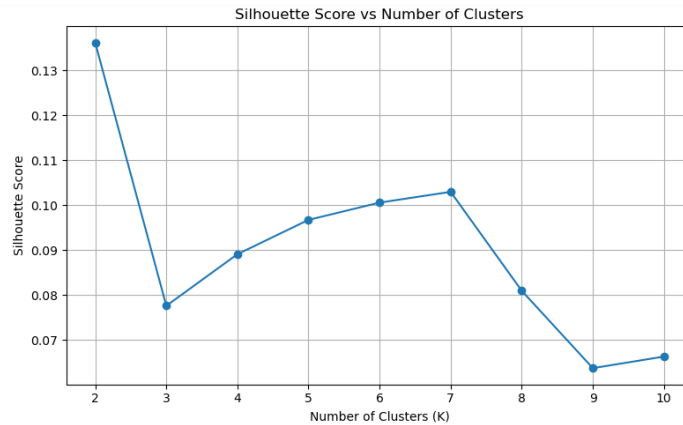
True Engagement Level	0	1	2
Cluster			
0	8287	15504	8202
1	2037	3870	2134

However, when comparing cluster assignments to the true engagement levels, the clusters still contain a mixture of different classes, suggesting that the data does not naturally separate into the predefined categories.

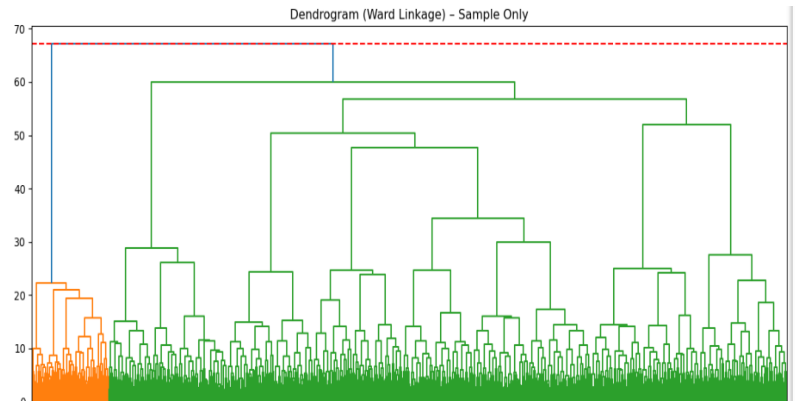
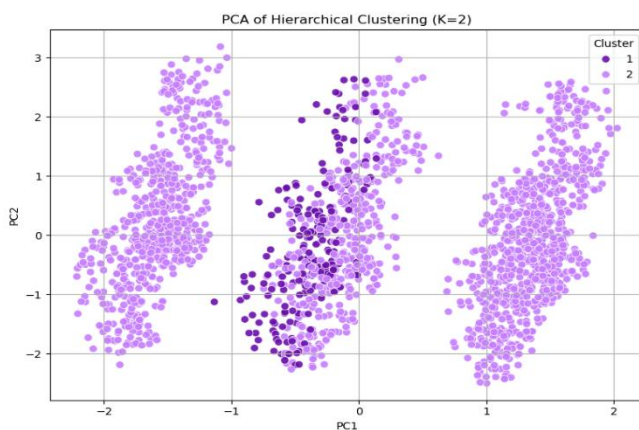
Overall, these results highlight the strong impact of feature selection on unsupervised learning, improving cluster separability even though perfect alignment with the target labels is not achieved.

Hierarchical Clustering- Model 10

Hierarchical clustering was applied using Ward linkage as an unsupervised learning method to further explore the structure of the data.



The optimal number of clusters was determined based on the silhouette score, with the best result obtained at $k = 2$ (silhouette score = 0.136).



The dendrogram provides a visual representation of the hierarchical relationships between observations and supports the selection of two main clusters.

A PCA-based visualization was used to examine cluster separation. While some grouping structure is observed, there is still noticeable overlap between clusters, indicating limited separation.

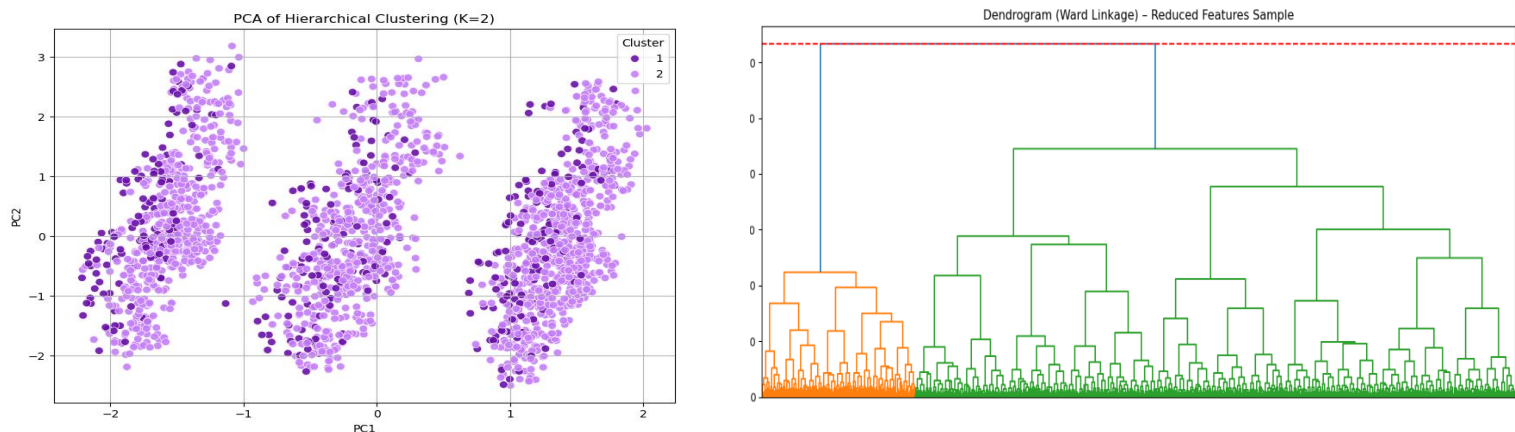
When comparing cluster assignments to the true engagement levels, each cluster contains a mixture of different classes, suggesting that the model does not fully capture the predefined categories.

Overall, the hierarchical clustering model identifies a basic structure in the data but provides limited alignment with the true engagement levels, similar to the results obtained with K-Means.

Hierarchical Clustering After Feature Filtering- Results

To further evaluate the impact of feature filtering, hierarchical clustering was re-applied using the reduced feature set derived from the logistic regression analysis.

The results show a significant improvement in clustering quality, with the silhouette score increasing from 0.136 to 0.275, while the optimal number of clusters remained $k = 2$.



The dendrogram and PCA visualization suggest a clearer separation between clusters compared to the model using all features, indicating that the reduced feature set enhances the structure of the data.

However, when comparing cluster assignments to the true engagement levels, each cluster still contains a mixture of different classes, suggesting that the data does not naturally separate into the predefined categories.

Overall, these findings confirm that feature filtering improves cluster separability in hierarchical clustering, even though full alignment with the target labels is not achieved

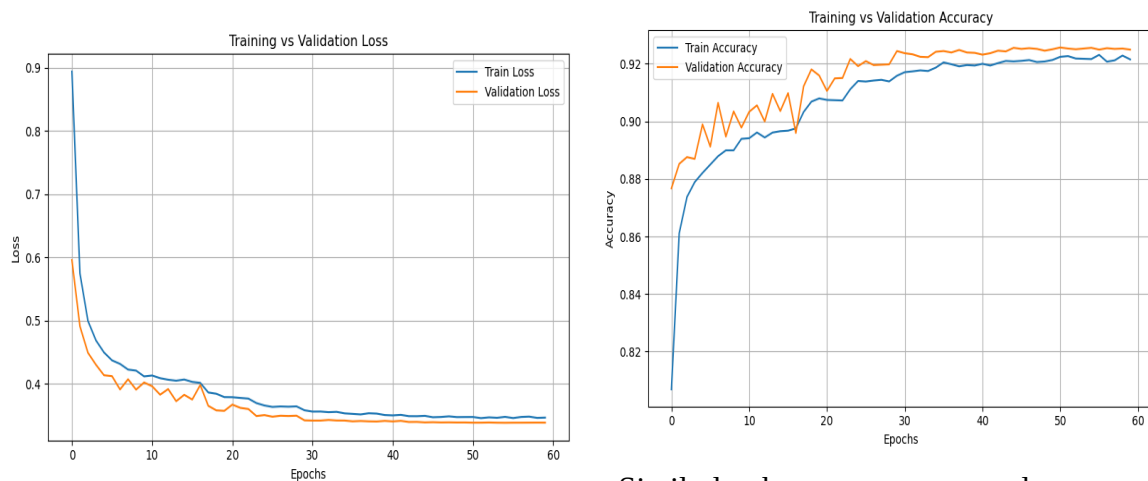
Neural Network- Model 11

The Neural Network model was implemented using a fully connected architecture and extensively tuned to optimize performance. Multiple configurations were tested, including variations in the number of layers, neurons, dropout layers, activation functions, regularization techniques, and optimization strategies.

The final architecture consisted of multiple dense layers ($800 \rightarrow 512 \rightarrow 256 \rightarrow 128 \rightarrow 64$ neurons) with ReLU activation, combined with dropout layers and L2 regularization to reduce overfitting. The output layer used a softmax activation function for multi-class classification.

The model was trained using the Adam optimizer with a learning rate of 0.001, along with callbacks such as Early Stopping and ReduceLROnPlateau to improve training efficiency and prevent overfitting.

To evaluate the training process of the neural network, learning curves for both loss and accuracy were analyzed across epochs. As shown in the figures, both training and validation loss decrease steadily and converge to similar values, indicating stable learning and effective optimization.



Similarly, the accuracy curves demonstrate a consistent improvement during training, with both training and validation accuracy converging around 92–93%. The close alignment between training and validation performance suggests that the model generalizes well and does not suffer from overfitting.

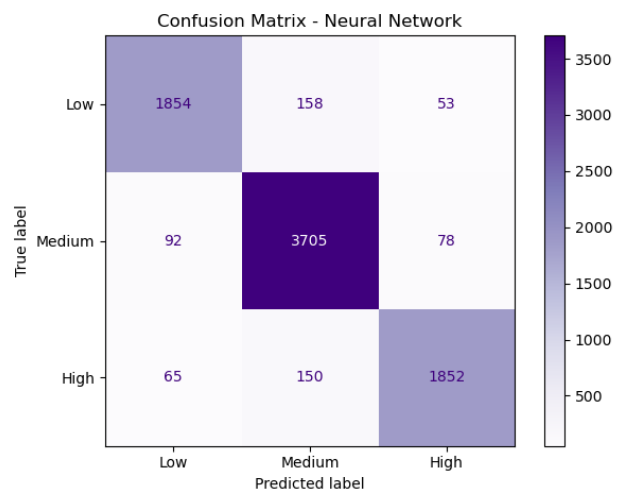
Overall, these results indicate that the chosen architecture and regularization techniques (Dropout and L2) successfully balance model complexity and generalization ability.

Results

	Precision	Recall	F1-score
0 (low)	0.92	0.90	0.91
1 (medium)	0.92	0.96	0.94
2 (high)	0.93	0.90	0.91
accuracy			0.926

The final model achieved a test accuracy of **0.926**, representing the highest performance among all models evaluated.

The confusion matrix shows that the neural network achieves high classification accuracy across all classes, with minimal misclassification. Most errors occur between adjacent levels, indicating that the model effectively captures the overall structure of the data.



ReduceLROnPlateau – Explanation

- * During the training process, the ReduceLROnPlateau callback was applied to dynamically adjust the learning rate. This method monitors the validation loss and reduces the learning rate when the model's performance stops improving. Specifically, if no improvement is observed for a predefined number of epochs (patience), the learning rate is multiplied by a given factor.
- * This approach helps the model converge more effectively by allowing larger updates in the early stages of training and smaller, more refined updates as the optimization process progresses. As a result, it improves stability and reduces the risk of getting stuck in suboptimal solutions.

Neural Network After Feature Filtering- Result

To further evaluate the impact of feature selection, the neural network model was re-trained using the reduced feature set derived from the logistic regression analysis.

The updated model achieved a training accuracy of 0.931 and a test accuracy of 0.929, resulting in a very small accuracy gap of 0.003.

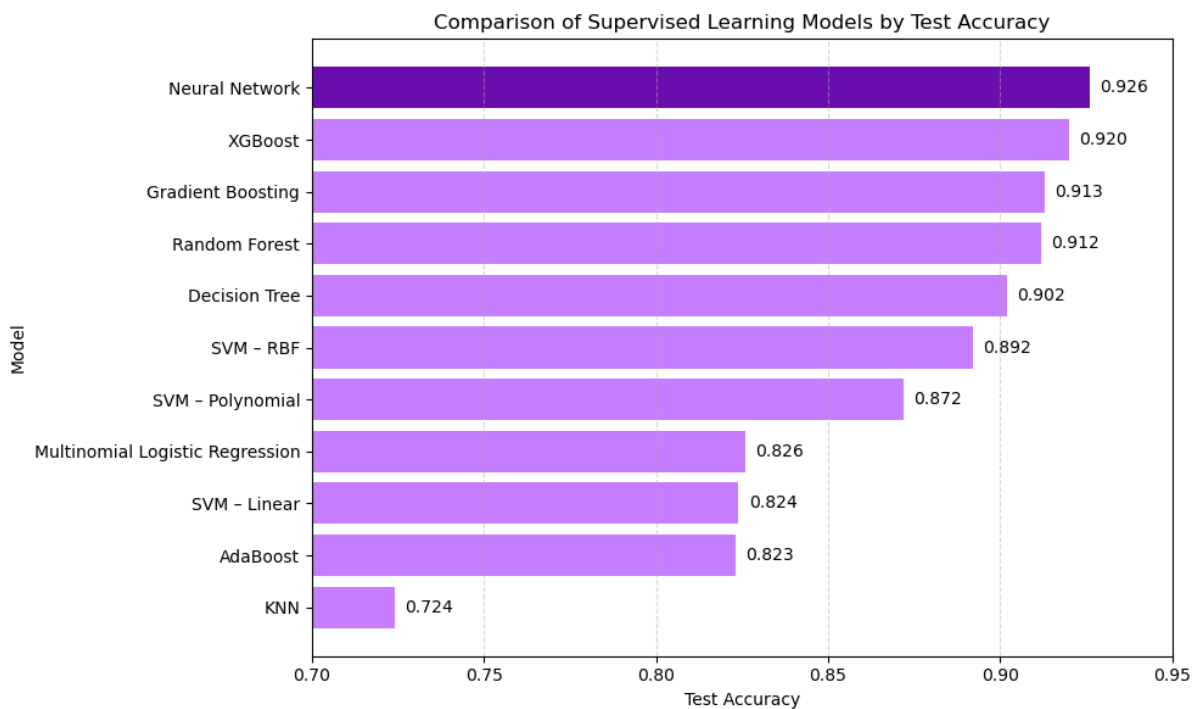
	Precision	Recall	F1-score
0 (low)	0.93	0.90	0.92
1 (medium)	0.92	0.96	0.94
2 (high)	0.94	0.90	0.92
accuracy			0.929

Overfitting check:

- **Using all features**, the model achieved a training accuracy of 0.933 and a test accuracy of 0.926, resulting in a small accuracy gap of approximately 0.007, indicating good generalization with no substantial overfitting.
- **After feature filtering**, the model achieved a training accuracy of 0.931 and an improved test accuracy of 0.929, with a very small accuracy gap of 0.003.

This suggests that feature filtering slightly improved predictive performance while further enhancing model generalization.

supervised models - Comparing results



Model	Accuracy-Test	Accuracy-Train	After Filtering Features (Test Accuracy)
Multinomial Logistic Regression	0.826	0.823	0.825
KNN	0.724	0.746	0.897
Decision Tree	0.902	0.920	0.904
Random Forest	0.912	1.00	0.921
Gradient Boosting	0.913	0.920	0.915
AdaBoost	0.823	0.821	0.822
XGBoost	0.920	0.936	0.921
SVM – Linear	0.824	0.825	0.825
SVM – Polynomial	0.872	0.892	0.881
SVM – RBF	0.892	0.906	0.922
Neural Network	0.926	0.933	0.929

Best-performing model: Neural Network. Achieving the highest test accuracy (0.926) while maintaining a very small train–test gap, indicating strong generalization and no significant overfitting.

Unsupervised models - Comparing results

Model	Best K	Silhouette Score	After Filtering	Improvement
K-Means	5 → 2	0.125 → 0.279	✓	Significant
Hierarchical	2	0.136 → 0.275	✓	Significant

The comparison of unsupervised models shows a clear improvement in clustering quality after feature filtering.

In both K-Means and Hierarchical Clustering, the silhouette score increased substantially, indicating better-defined and more separated clusters. In addition, the optimal number of clusters became more stable and interpretable ($k = 2$), suggesting a clearer underlying structure in the data.

However, despite the improvement in cluster separation, the resulting clusters still do not perfectly align with the predefined engagement levels, indicating that the data does not naturally separate into these categories.

Conclusion and Model Comparison

This project applied multiple machine learning models to classify user engagement levels, including traditional algorithms, ensemble methods, and a neural network approach.

Overall, most models achieved relatively high predictive performance when trained on the full set of features. In particular, ensemble methods such as Random Forest, Gradient Boosting, and XGBoost demonstrated strong accuracy, highlighting their effectiveness in handling structured datasets. The neural network also showed consistently high performance, with stable learning behavior and minimal overfitting.

However, a key finding of this project is the significant impact of feature selection. After removing statistically non-significant variables based on the multinomial logistic regression analysis, most models showed improvements in performance and generalization. In several cases, the improvement was substantial, such as in KNN and clustering models, where both accuracy and structural clarity increased notably.

In supervised models, feature filtering led to higher test accuracy and reduced overfitting, as reflected by smaller gaps between training and test performance. In unsupervised models (K-Means and Hierarchical Clustering), the silhouette score improved significantly, indicating better separation and a clearer underlying data structure.

An exception to this pattern was observed in the AdaBoost model, where feature filtering did not lead to an improvement. The model trained on the full feature set achieved a slightly higher test accuracy compared to the filtered version, although the difference was minimal. This suggests that AdaBoost may be less sensitive to irrelevant features due to its boosting mechanism, which focuses on difficult observations during training.

Among all models, the neural network consistently demonstrated the best overall performance. It achieved the highest accuracy while maintaining a very small train-test gap, indicating strong generalization. The use of regularization techniques, including Dropout and L2 penalties, along with adaptive learning rate optimization, contributed to a stable and efficient training process.

Ensemble methods such as XGBoost and Gradient Boosting also performed competitively, confirming their robustness and reliability. However, the neural network showed superior capability in capturing complex relationships within the data.

In conclusion, while multiple models are suitable for this classification task, the neural network emerges as the most reliable and effective model. Furthermore, this project highlights the importance of feature selection as a critical step in improving both predictive performance and model stability across different machine learning approaches.